

UID EXPERIMENT 11

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Create a data visualization (e.g., pie charts, bar graphs) for an library management system using javascript.

AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for an library management system using JavaScript.

PROCEDURE:

Step 1: Set Up Your HTML File

First, create an HTML file to hold your canvas for the chart and include Chart.js.

Html

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <title>Library Management Dashboard</title>
```

```
<!-- Include Chart.js -->
```

```
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
```

```
<style>
```

```
  body {  
    font-family: Arial, sans-serif;  
    background: #f5f5f5;  
    text-align: center;  
  }
```

```
  h1 {  
    margin-top: 20px;  
  }
```

```
  .chart-container {  
    width: 60%;  
    margin: 30px auto;  
  }
```

```
  canvas {  
    background: white;  
    padding: 15px;  
    border-radius: 10px;  
  }
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1>Library Management System Dashboard</h1>
```

```
<!-- Pie Chart -->
```

```
<div class="chart-container">
```

```
  <h3>Books by Category</h3>
```

```
  <canvas id="categoryChart"></canvas>
```

```
</div>
```

```
<!-- Bar Chart -->
```

```
<div class="chart-container">
```

```
  <h3>Books Issued This Week</h3>
```

```
  <canvas id="issuedChart"></canvas>
```

```
</div>
```

```
<script>
```

```
  // Pie Chart - Book Categories
```

```
  const categoryChart = new Chart(document.getElementById('categoryChart'),
  {
    type: 'pie',
    data: {
      labels: ['Fiction', 'Non-Fiction', 'Science', 'History', 'Comics'],
      datasets: [{
        data: [30, 20, 25, 15, 10],
        backgroundColor: ['red', 'blue', 'green', 'orange', 'purple']
      }]
    }
  })
```

```
    }  
  });
```

```
// Bar Chart - Books Issued Per Day
```

```
const issuedChart = new Chart(document.getElementById('issuedChart'), {  
  type: 'bar',  
  data: {  
    labels: ['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat'],  
    datasets: [{  
      label: 'Books Issued',  
      data: [12, 19, 8, 15, 10, 6],  
      backgroundColor: 'teal'  
    }]  
  }  
});
```

```
</script>
```

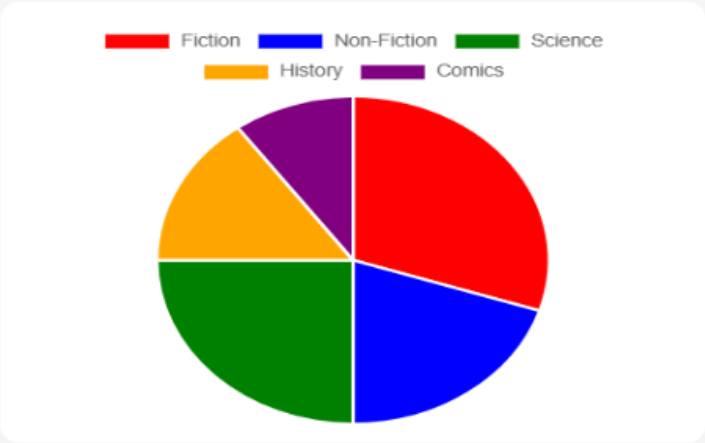
```
</body>
```

```
</html>
```

OUTPUT:

Library Management System Dashboard

Books by Category



Books Issued This Week

